

ROSHAN KUMAR

B. Tech Computer Science And Engineering - IIIT Kalyani

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🌐 LinkedIn 🐙 Github 📁 Portfolio

Education

Indian Institute of Information Technology, Kalyani

2021 – 2025

Bachelor of Technology - Computer Science and Engineering, CGPA: 7.9

Kalyani, West Bengal

Experience

Meril Life Sciences Pvt. Ltd. (NuvoAi) - SDE - 1

June 2025 - Present

AI Chat & RAG Pipeline Engineering

On-site

- Developed an end-to-end **RAG pipeline** using Milvus for vector similarity search (cosine metric) with configurable reranking (0.3 threshold) achieving high-precision semantic retrieval.
- Built **multi-modal AI search agents** enabling web search, academic search (PubMed, Google Scholar), WolframAlpha computation, YouTube/Reddit content extraction, and multi-engine image search.
- Implemented a **document processing pipeline** supporting PDF, DOCX, and TXT uploads using RecursiveCharacterTextSplitter (500-token chunks, 100-token overlap) with embeddings stored in MinIO.
- Integrated multiple **Ollama LLM models** with SSE-based streaming responses enabling dynamic model selection for chat, RAG, and embedding workflows.
- Designed secure **authentication and authorization** using JWT, bcrypt hashing, and sensitive-query keyword filtering for safe LLM operations.

NuvoAi (Merai) - Full Stack Developer Intern — [Certificate](#)

Jan 2025 - June 2025

Real-Time Search System and Gene Dataset Platform

On-site

- Search Engine Backend (Nuomics Project):** Developed a scalable backend search engine using Node.js and Elasticsearch for diseases, genes, and datasets with multi-index querying logic.
- Built RESTful APIs with advanced filtering, sorting, and highlighting; optimized Elasticsearch queries for large-scale biomedical datasets.
- Real-Time Search Infrastructure:** Implemented debounced and paginated search APIs ensuring minimal latency; used query DSL and aggregations for intelligent retrieval.
- Gene/Dataset Information Service:** Built backend routes to fetch detailed gene/dataset information from Elasticsearch enabling dynamic UI rendering.
- Ensured seamless frontend-backend integration using Express.js with robust Elasticsearch → API → UI data pipelines.

Projects

AI YouTube RAG Assistant — [Github](#)

- Built an **AI-powered YouTube RAG assistant** enabling **advanced Q&A, fact-checking, and timestamped summaries**.
- Integrated **Ollama + Llama-3** with **hybrid RAG** combining **video context + external web knowledge**.
- Implemented **conversation memory, follow-up question handling, and semantic question classification**.
- Optimized retrieval using **FAISS, HuggingFace embeddings, window retrieval, and dynamic K selection**.
- Technologies:** FastAPI, Python, Ollama, Llama-3, LangChain, FAISS, HuggingFace Embeddings, DuckDuckGo Search, YouTube API, Docker, Docker Compose

CodingWeb - Ed-Tech Platform Using MERN Stack — [Github](#) — [Live](#)

- Developed a **MERN stack** platform for creating, consuming, and rating educational content with **user authentication, course management, payment integration, and Cloudinary media management**.
- Implemented **responsive UIs, RESTful APIs, Redux** state management, and secure **JWT authentication**. Deployed using **Vercel, Render, and MongoDB Atlas**.

Skills

Programming Languages: C, C++, HTML, CSS, JavaScript, Python

Technologies & Frameworks: React, Node.js, Express.js, TailwindCSS, Git, GitHub, Django, FastApi, Langchain

Databases: MySQL, MongoDB, Docker, Elasticsearch

Core Subjects: Operating Systems, OOP, DBMS, Data Structures & Algorithms

Sports Programming

- Leetcode:** Username: [Acoder0001](#)
- GeeksforGeeks:** Username: [roshan9888](#)

Achievements

- Qualified for Level 2 among top 1000 out of 484,306 participants in **Flipkart GRID 6.0**
- Solved over 900+ problems on LeetCode ([Acoder0001](#)) and GeeksforGeeks ([roshan9888](#))
- Ranked 5th in intra-college CodeRush Challenge at IIIT Kalyani by AlgoUniversity (Junior Topper)
- Contributed to TNP IIIT Kalyani placement portal enhancing functionality and user experience
- Completed Basic SQL Test on HackerRank [Certificate](#) and Node.js Essential Training [Certificate](#)